

Data Engineering Learning Path

A Complete Roadmap to Becoming a Data Engineer

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Introduction

Data Engineering is one of the fastest-growing fields in tech. Data Engineers build and maintain the infrastructure that enables data scientists and analysts to do their work. This roadmap will take you from foundational skills to job-ready expertise in approximately 10-12 months.

What Does a Data Engineer Do?

- Build and maintain data pipelines (ETL/ELT)

- Design and manage data warehouses and data lakes
- Ensure data quality and reliability
- Optimize data storage and retrieval
- Work with big data technologies and cloud platforms
- Collaborate with data scientists, analysts, and stakeholders

Estimated Timeline

- **Total Duration:** 10-12 months of consistent study
 - **Time Commitment:** 15-20 hours per week
 - **Intensity:** Mix of theory (40%) and hands-on practice (60%)
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Current Skills Assessment

Skills You Already Have

- **Python:** Programming fundamentals
- **SQL:** Database querying
- **HTML/CSS/JavaScript:** Web development basics
- **Git/GitHub:** Version control

Skills to Acquire

- Advanced SQL and database design
 - ETL/ELT pipeline development
 - Workflow orchestration (Apache Airflow)
 - Cloud platforms (AWS/GCP/Azure)
 - Big data processing (Apache Spark)
 - Containerization (Docker)
 - Data modeling and warehousing
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Phase 1: Strengthen Foundations (2-3 months)

1.1 Advanced SQL & Databases

What to Learn: - **Window Functions:** ROW_NUMBER(), RANK(), LEAD(), LAG() -

Common Table Expressions (CTEs): Recursive and non-recursive - **Query**

Optimization: EXPLAIN plans, indexing strategies - **Advanced Joins:** Self-joins, cross joins, lateral joins - **Stored Procedures & Functions:** PL/pgSQL - **Transactions &**

ACID: Isolation levels, locking - **Database Design:** Normalization (1NF to BCNF), denormalization

Hands-On Practice: - Solve SQL problems on LeetCode, HackerRank, SQLZoo - Work with a real PostgreSQL database - Optimize slow queries - Design schemas for real-world scenarios

Resources: - PostgreSQL Documentation - "SQL Performance Explained" by Markus Winand - Mode Analytics SQL Tutorial - PostgreSQL Exercises (pgexercises.com)

Time Estimate: 4-5 weeks

1.2 Python for Data Engineering

What to Learn: - **Pandas & NumPy:** DataFrames, Series, data manipulation - **Data**

Validation: Pydantic, Great Expectations - **Working with APIs:** requests, authentication, pagination - **File Formats:** CSV, JSON, XML, Parquet, Avro, ORC -

Error Handling: try/except, logging - **Testing:** pytest, unittest - **Virtual Environments:** venv, conda - **Package Management:** pip, requirements.txt

Hands-On Practice: - Read data from various file formats - Clean and transform messy datasets - Consume REST APIs and process responses - Write unit tests for data transformation functions - Build a data validation pipeline

Resources: - "Python for Data Analysis" by Wes McKinney - Real Python tutorials - Pandas Documentation - Kaggle Python courses

Time Estimate: 3-4 weeks

1.3 Linux & Command Line

What to Learn: - **Basic Commands:** ls, cd, mkdir, cp, mv, rm - **File Manipulation:** grep, awk, sed, cut, sort, uniq - **Process Management:** ps, top, kill - **Permissions:** chmod, chown - **SSH:** Connecting to remote servers - **Cron Jobs:** Scheduling tasks - **Shell Scripting:** Bash basics, variables, loops

Hands-On Practice: - Set up a Linux VM or WSL - Write bash scripts for data processing - Schedule automated tasks with cron - Process log files with command-line tools

Resources: - Linux Journey (linuxjourney.com) - "The Linux Command Line" by William Shotts - OverTheWire Bandit ([wargames](https://overthewire.org/bandit/))

Time Estimate: 2-3 weeks

1.4 Introduction to NoSQL

What to Learn: - NoSQL vs SQL databases - Document stores: MongoDB - Key-value stores: Redis - Column-family stores: Cassandra (overview) - When to use NoSQL vs SQL

Hands-On Practice: - Install and use MongoDB - Perform CRUD operations - Understand document schema design - Compare with relational databases

Time Estimate: 2 weeks

Phase 2: Core Data Engineering (3-4 months)

2.1 ETL/ELT Fundamentals

What to Learn: - **ETL vs ELT:** When to use each approach - **Data Extraction:** From APIs, databases, files, web scraping - **Data Transformation:** Cleaning, aggregation, enrichment - **Data Loading:** Bulk loading, incremental loading, upserts - **Idempotency:** Ensuring pipeline reruns are safe - **Error Handling:** Retries, dead letter queues

Hands-On Practice: - Build Python ETL scripts - Extract data from multiple sources - Transform and clean real-world messy data - Load data into PostgreSQL - Handle failures gracefully

Project Ideas: - COVID-19 data tracker pipeline - Stock market data aggregator - E-commerce sales data pipeline - Weather data collector

Time Estimate: 4-5 weeks

2.2 Workflow Orchestration with Apache Airflow

What to Learn: - **DAGs (Directed Acyclic Graphs):** Defining workflows - **Operators:** PythonOperator, BashOperator, PostgresOperator - **Task Dependencies:** set_upstream, set_downstream, >> - **Scheduling:** Cron expressions, intervals - **XComs:** Passing data between tasks - **Sensors:** Waiting for conditions - **Hooks:** Connecting to external systems - **Monitoring:** UI, logs, alerts

Hands-On Practice: - Install Airflow locally with Docker - Create your first DAG - Schedule ETL pipelines - Set up task dependencies - Monitor and debug failed tasks - Implement retries and SLAs

Resources: - Apache Airflow Documentation - "Data Pipelines with Apache Airflow" by Bas Harens - Astronomer Academy (free courses) - Marc Lamberti's Airflow courses

Time Estimate: 5-6 weeks

2.3 Data Modeling

What to Learn: - **Dimensional Modeling:** Star schema, snowflake schema - **Fact Tables:** Additive, semi-additive, non-additive measures - **Dimension Tables:** Slowly Changing Dimensions (SCD Type 1, 2, 3) - **Normalization vs Denormalization:** Trade-offs - **Surrogate Keys:** When and why to use them - **Data Vault:** Advanced modeling technique

Hands-On Practice: - Design a star schema for a business scenario - Implement SCD Type 2 dimensions - Build a small data warehouse - Write queries against dimensional models

Resources: - "The Data Warehouse Toolkit" by Ralph Kimball - Kimball Group website - Data modeling tutorials on YouTube

Time Estimate: 3-4 weeks

2.4 Docker & Containerization

What to Learn: - **Docker Basics:** Images, containers, volumes - **Dockerfile:** Creating custom images - **Docker Compose:** Multi-container applications - **Networking:** Container communication - **Volumes:** Persistent data storage - **Best Practices:** Small images, layer caching

Hands-On Practice: - Install Docker Desktop - Run PostgreSQL in a container - Containerize a Python ETL script - Use Docker Compose for Airflow - Build and push images to Docker Hub

Resources: - Docker Documentation - Docker for Beginners (YouTube) - Play with Docker (labs.play-with-docker.com)

Time Estimate: 3-4 weeks

Phase 3: Big Data & Cloud (3-4 months)

3.1 Data Warehousing

What to Learn: - **Data Warehouse Concepts:** OLAP vs OLTP - **dbt (data build tool):** Transformations as code - **Partitioning:** Date-based, range, list partitioning - **Clustering:** Optimizing query performance - **Materialized Views:** Pre-computed results - **Modern Data Stack:** ELT approach

Cloud Data Warehouses: - **Snowflake:** Multi-cluster architecture - **Google BigQuery:** Serverless, columnar storage - **Amazon Redshift:** MPP architecture

Hands-On Practice: - Set up dbt with PostgreSQL - Write dbt models and tests - Implement incremental models - Create documentation with dbt docs - Try BigQuery or Snowflake free tier

Resources: - dbt Documentation and Courses - Snowflake University (free) - Google Cloud Skills Boost (BigQuery) - "The Data Warehouse Toolkit" by Kimball

Time Estimate: 4-5 weeks

3.2 Big Data Processing with Apache Spark

What to Learn: - **Distributed Computing:** Concepts and architecture - **PySpark:** Python API for Spark - **RDDs:** Resilient Distributed Datasets - **DataFrames & Datasets:** Structured APIs - **Transformations:** map, filter, groupBy, join - **Actions:** collect, count, save - **Lazy Evaluation:** Execution plans - **Performance Optimization:** Partitioning, caching, broadcast joins

Hands-On Practice: - Install Spark locally or use Databricks Community Edition - Process large CSV/JSON files - Perform aggregations and joins - Read/write Parquet files - Optimize Spark jobs

Resources: - Apache Spark Documentation - Databricks Academy (free courses) - "Learning Spark" by Jules Damji et al. - Spark tutorials on DataCamp

Time Estimate: 5-6 weeks

3.3 Cloud Platforms (Choose ONE to Start)

Option A: Amazon Web Services (AWS)

Core Services to Learn: - **S3:** Object storage for data lakes - **RDS:** Managed relational databases - **Redshift:** Data warehouse - **Glue:** ETL service and data catalog - **Lambda:** Serverless functions - **Kinesis:** Real-time data streaming - **IAM:** Security and permissions - **CloudWatch:** Monitoring and logging

Hands-On Practice: - Create S3 buckets and upload data - Set up RDS PostgreSQL instance - Build ETL job with AWS Glue - Create Lambda function for data processing - Configure IAM roles and policies

Certification: AWS Certified Data Analytics - Specialty

Option B: Google Cloud Platform (GCP)

Core Services to Learn: - **Cloud Storage:** Object storage - **BigQuery:** Serverless data warehouse - **Cloud SQL:** Managed databases - **Dataflow:** Stream and batch processing - **Pub/Sub:** Messaging service - **Cloud Functions:** Serverless - **IAM:** Access control - **Cloud Logging:** Monitoring

Hands-On Practice: - Upload data to Cloud Storage - Query data in BigQuery - Create Dataflow pipeline - Set up Pub/Sub topics

Certification: Google Professional Data Engineer

Option C: Microsoft Azure

Core Services to Learn: - **Blob Storage:** Object storage - **Azure SQL Database:** Managed SQL - **Synapse Analytics:** Data warehouse - **Data Factory:** ETL/ELT service - **Databricks:** Spark platform - **Event Hubs:** Streaming - **Azure Functions:** Serverless

Time Estimate for Cloud: 6-8 weeks

3.4 Data Streaming Basics

What to Learn: - **Stream Processing Concepts:** Real-time vs batch - **Apache Kafka:** Distributed event streaming - **Topics & Partitions:** Data organization - **Producers & Consumers:** Publishing and reading - **Message Queues:** RabbitMQ, AWS SQS - **Use Cases:** Real-time analytics, event-driven architecture

Hands-On Practice: - Install Kafka with Docker - Create topics and partitions - Write producers and consumers in Python - Process streaming data - Integrate with databases

Resources: - Kafka Documentation - Confluent Developer courses - "Kafka: The Definitive Guide" by Neha Narkhede

Time Estimate: 4-5 weeks

Phase 4: Advanced Topics (2-3 months)

4.1 Data Quality & Testing

What to Learn: - **Data Validation:** Schema validation, business rules - **Great Expectations:** Data quality framework - **Unit Testing:** Testing transformations - **Integration Testing:** End-to-end pipeline tests - **Data Lineage:** Tracking data flow - **Monitoring:** Alerts for data issues - **SLAs:** Service level agreements

Tools: - Great Expectations - dbt tests - pytest - Apache Griffin

Time Estimate: 3-4 weeks

4.2 Infrastructure as Code

What to Learn: - **Terraform:** Provisioning cloud infrastructure - **Resources:** Defining infrastructure components - **State Management:** Tracking infrastructure - **Modules:** Reusable components - **Version Control:** Git for infrastructure

Hands-On Practice: - Write Terraform for AWS/GCP resources - Provision databases and storage - Manage infrastructure changes - Destroy and recreate environments

Time Estimate: 3-4 weeks

4.3 CI/CD for Data Pipelines

What to Learn: - **Continuous Integration:** Automated testing - **Continuous Deployment:** Automated deployment - **GitHub Actions / GitLab CI:** Pipeline automation - **Testing Strategies:** Unit, integration, smoke tests - **Environment Management:** Dev, staging, production

Hands-On Practice: - Set up GitHub Actions workflows - Automate testing and deployment - Implement blue-green deployments

Time Estimate: 2-3 weeks

4.4 Performance & Optimization

What to Learn: - **Query Optimization:** Indexes, query plans - **Pipeline Optimization:** Parallelization, incremental processing - **Cost Optimization:** Cloud cost management - **Caching:** Redis, in-memory caching - **Compression:** Parquet, Snappy, Gzip

Time Estimate: 2-3 weeks

Project-Based Milestones

Build these projects progressively to demonstrate your skills:

Month 1-2: Basic ETL Project

Project: Web Scraper with Database Storage - Scrape data from a website (e.g., news, weather, stocks) - Clean and transform the data - Store in PostgreSQL - Automate with cron or Python schedule

Skills Demonstrated: Python, SQL, data extraction, automation

Month 3-4: Orchestrated Pipeline

Project: Multi-Source ETL with Airflow - Extract from 3+ sources (API, CSV, database) - Transform and validate data - Load to PostgreSQL - Orchestrate with Airflow - Monitor and log

Skills Demonstrated: ETL, Airflow, data validation, orchestration

Month 5-6: Data Warehouse

Project: Analytics Data Warehouse with dbt - Build star schema with fact and dimension tables - Implement SCD Type 2 - Use dbt for transformations - Create analytics queries - Generate dbt documentation

Skills Demonstrated: Data modeling, dbt, dimensional modeling, analytics

Month 7-8: Big Data Processing

Project: Large-Scale Data Processing with Spark - Process millions of records - Perform complex aggregations - Join multiple large datasets - Optimize performance - Save results in Parquet format

Skills Demonstrated: PySpark, distributed computing, performance optimization

Month 9-10: Cloud & Streaming

Project: Real-Time Data Pipeline in Cloud - Ingest real-time data (Kafka or cloud streaming service) - Process with Spark Streaming or cloud service - Store in cloud data warehouse - Build dashboard for real-time metrics

Skills Demonstrated: Streaming, cloud platforms, real-time processing

Month 11-12: Production-Ready System

Project: End-to-End Production Pipeline - Full ETL/ELT pipeline - Infrastructure as code (Terraform) - CI/CD with automated testing - Monitoring and alerting - Data quality checks - Documentation

Skills Demonstrated: Everything! Production-ready, professional-grade system

Learning Resources

Online Courses

Comprehensive Platforms: - **DataCamp:** Data Engineer with Python track - **Udemy:** - "The Complete Apache Airflow Course" - "Apache Spark with Python" - "AWS Certified Data Analytics" - **Coursera:** - Google Cloud Data Engineering - IBM Data Engineering Professional Certificate - **LinkedIn Learning:** Various data engineering courses

Specialized Learning: - **Astronomer Academy:** Airflow (free) - **Databricks Academy:** Spark (free) - **Snowflake University:** Data warehousing (free) - **Google Cloud Skills Boost:** GCP services (free tier) - **AWS Skill Builder:** AWS services (free tier)

Books

Essential Reading: 1. **"Fundamentals of Data Engineering"** by Joe Reis & Matt Housley - Modern best practices and concepts

1. **"Designing Data-Intensive Applications"** by Martin Kleppmann
2. Deep dive into distributed systems
3. **"The Data Warehouse Toolkit"** by Ralph Kimball
4. Dimensional modeling bible
5. **"Data Pipelines with Apache Airflow"** by Bas Harenslak & Julian de Ruiter
6. Comprehensive Airflow guide
7. **"Learning Spark"** by Jules Damji et al.
8. Spark fundamentals and advanced topics

Advanced Reading: 6. "SQL Performance Explained" by Markus Winand 7. "Streaming Systems" by Tyler Akidau et al. 8. "Python for Data Analysis" by Wes McKinney

YouTube Channels & Creators

- **Seattle Data Guy:** Data engineering career advice and tutorials
- **Data Engineering Zoomcamp** (DataTalks.Club): Free comprehensive course
- **Karolina Sowinska:** Data engineering projects
- **Andreas Kretz:** DE Academy channel
- **TechWorld with Nana:** DevOps and cloud
- **freeCodeCamp:** Various data engineering courses

Practice Platforms

Coding Practice: - **LeetCode**: SQL and algorithm problems - **HackerRank**: SQL, Python challenges - **SQLZoo**: Interactive SQL tutorials - **Mode Analytics**: SQL practice with real datasets - **Kaggle**: Datasets and competitions

Cloud Practice: - **AWS Free Tier**: 12 months free - **GCP Free Tier**: \$300 credit - **Azure Free Tier**: \$200 credit - **Databricks Community Edition**: Free Spark environment

Communities & Forums

- **r/dataengineering** (Reddit): Active community
 - **Data Engineering Discord servers**
 - **Stack Overflow**: Q&A
 - **LinkedIn Groups**: Data Engineering professionals
 - **Local Meetups**: Check Meetup.com for data engineering groups
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Blogs & Websites

- **DataEngineeringWeekly.com**: Newsletter
 - **Locally Optimistic**: Data blog
 - **The Data Engineering Podcast**
 - **dbt Blog**: Modern data stack insights
 - **AWS Big Data Blog**
 - **Google Cloud Blog**
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Key Skills Employers Want

Must-Have Skills (Core)

Programming & Databases: -  SQL (Advanced) - **CRITICAL** -  Python - **CRITICAL** - Version Control (Git) - **CRITICAL**

Data Engineering Tools: - Apache Airflow or similar orchestration - **CRITICAL** - ETL/ELT experience - **CRITICAL** - Docker & containerization - **HIGHLY DESIRED**

Cloud Platforms: - AWS, GCP, or Azure (at least one) - **CRITICAL** - Understanding of cloud data services - **CRITICAL**

Nice-to-Have Skills (Bonus)

Big Data: - Apache Spark/PySpark - Hadoop ecosystem (basic understanding) - Distributed computing concepts

Streaming: - Apache Kafka - Real-time processing - Event-driven architecture

Modern Data Stack: - dbt (data build tool) - Snowflake/BigQuery/Redshift - Fivetran/Airbyte (ELT tools)

DevOps: - Terraform or CloudFormation - CI/CD pipelines - Kubernetes (basic understanding)

Data Quality: - Great Expectations - Data validation frameworks - Testing practices

Soft Skills (Often Overlooked!)

- **Communication:** Explain technical concepts to non-technical stakeholders
- **Problem-Solving:** Debug complex data issues
- **Collaboration:** Work with data scientists, analysts, engineers
- **Documentation:** Write clear docs for pipelines and systems
- **Business Acumen:** Understand business requirements

Weekly Study Schedule

Template Schedule (15-20 hours/week)

Monday (2-3 hours): - 🎓 Theory & Concepts - Watch tutorials or read documentation - Take notes, understand fundamentals

Tuesday (2-3 hours): - 🎓 Theory & Concepts (continued) - Deep dive into specific topics - Complete online course modules

Wednesday (2-3 hours): - 💻 Hands-On Practice - Code along with tutorials - Practice exercises on platforms (LeetCode, DataCamp)

Thursday (2-3 hours): - 💻 Hands-On Practice - Build small proof-of-concept projects - Experiment with new tools

Friday (2-3 hours): - 💻 Hands-On Practice - Debug and troubleshoot - Review week's learning

Saturday (3-4 hours): - 🚀 Portfolio Project Work - Apply week's learning to main project - Build, test, iterate

Sunday (2-3 hours): - 🚀 Portfolio Project Work (continued) - Document your code - Write README, add to GitHub - **Optional:** Review and plan next week

Study Tips for Success

1. Build in Public: - Share your learning journey on LinkedIn - Post projects on GitHub - Write blog posts about what you're learning

2. Follow the 80/20 Rule: - 20% theory, 80% hands-on practice - Learning by doing is most effective

3. Don't Get Stuck in Tutorial Hell: - After learning a concept, build something without following a tutorial - Struggle is where learning happens

4. Join Communities: - Ask questions on Stack Overflow, Reddit - Attend virtual meetups - Network with other learners and professionals

5. Track Your Progress: - Keep a learning journal - Celebrate small wins - Review and reflect regularly

6. Focus on Fundamentals First: - Master SQL and Python before moving to advanced tools - Strong foundations make everything else easier

7. Don't Learn Everything: - You don't need to know every tool - Focus on core skills, specialize later

Career Roadmap

Entry-Level Roles to Target

Junior Data Engineer: - Build and maintain ETL pipelines - Write SQL queries - Basic Python scripting - Support senior engineers

Data Analyst (Transition Role): - Strong SQL skills - Data analysis and reporting - Can transition to DE role

ETL Developer: - Focus on data integration - Transform and load data - Pipeline development

BI Developer (Transition Role): - Data warehousing - Dimensional modeling - Can pivot to data engineering

Typical Salary Ranges (US, 2025)

- **Junior Data Engineer:** \$70,000 - \$95,000
- **Mid-Level Data Engineer:** \$95,000 - \$130,000
- **Senior Data Engineer:** \$130,000 - \$180,000
- **Staff/Principal Data Engineer:** \$180,000 - \$250,000+

Note: Varies significantly by location, company, and experience

How to Get Your First Job

- 1. Build a Strong Portfolio:** - 3-5 projects showcasing different skills - Clean, well-documented code on GitHub - README files explaining projects - Live demos or screenshots
 - 2. Create a Data Engineering Resume:** - Highlight relevant projects - List technical skills clearly - Quantify achievements (e.g., "Built pipeline processing 1M+ records daily") - Include links to GitHub and portfolio
 - 3. Network:** - Connect with data engineers on LinkedIn - Attend meetups and conferences - Engage with the community online
 - 4. Apply Strategically:** - Target junior/entry-level roles - Consider startups (more willing to take juniors) - Look for "Associate" or "Junior" positions - Consider internships or contract roles
 - 5. Prepare for Interviews:** - **SQL:** Practice complex queries - **Python:** Data structures, algorithms - **System Design:** Design a data pipeline - **Behavioral:** STAR method responses - **Projects:** Be ready to deep-dive into your work
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Conclusion

Becoming a data engineer is a rewarding journey that requires dedication and consistent practice. With your existing Python and SQL skills, you already have a strong foundation. Follow this roadmap, build projects, and stay curious.

Remember: - Learning never stops in data engineering - Technologies evolve rapidly - Fundamentals remain constant - Hands-on practice is key - Community support accelerates learning

Your Next Steps: 1. Save this learning path 2. Choose Phase 1 topic to start with 3. Set up your learning environment 4. Build your first project 5. Share your progress

Good luck on your data engineering journey!

Additional Resources

GitHub Repositories to Study

- **Awesome Data Engineering**: Curated list of resources
- **Data Engineering Cookbook**: Recipes for common tasks
- **DataTalks.Club**: Data Engineering Zoomcamp materials

Podcasts

- **Data Engineering Podcast** by Tobias Macey
- **The Data Stack Show**
- **Analytics Engineering Podcast**

Newsletters

- **Data Engineering Weekly**
- **Seattle Data Guy Newsletter**
- **dbt Newsletter**

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